

14. No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	SWTID-2026-8135
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

S.No	Feature Name	Feature Description	Module / Component	Status
1	AI Q&A Chat	Answers free-form educational questions with a structured answer, explanation, example and key points	qna.py — /api/ask, /qa	Done
2	Concept Explanation	Explains a concept at simple / medium / advanced level, using a local LaMini-Flan-T5 model with Gemini fallback	explanation_module.py — /api/explain, /explain/	Done
3	MCQ Quiz Generation	Generates 1–15 multiple-choice questions from a topic or pasted text, each with an explanation	quiz_module.py — /api/quiz, /quiz	Done
4	Text / Notes Summarization	Produces a summary, bullet points and keywords from pasted text	summary_module.py — /api/summary, /summarize/	Done
5	Personalized Learning Roadmap	Generates learning resources (book/course/practice) and a step-by-step roadmap for a topic and level	learning_path.py — /api/recommend, /learn/recommendations	Done
6	Demo / Mock Mode	Serves realistic mock responses for every feature when no Gemini API key is configured	utils/gemini_service.py	Done
7	Query / Response Logging	Logs every query and its AI response to a JSON data store for traceability	utils/db_manager.py	Done
8	Responsive Multi-Page Frontend	Eight pages (home, chat, explain, quiz, summary, recommend, about, contact) with a light/dark theme toggle	templates/, static/css, static/js	Done

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (Chat, Explain, Quiz, Summary, Recommend)
Additional / Nice-to-Have Features	3 (Demo mode, JSON logging, responsive theming)
Features Tested & Verified	8 (all covered directly or indirectly by tests/test_api.py)

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	8 pages	Chat dashboard, Quiz dashboard with timer/scoring, theme toggle
2	Backend / Logic	5 core modules	qna.py, explanation_module.py, quiz_module.py, summary_module.py, learning_path.py
3	Database / Storage	6 JSON collections	queries.json, responses.json, quizzes.json, summaries.json, learning_paths.json, users.json
4	API / Integration	2 AI providers	Google Gemini API, local HuggingFace LaMini-Flan-T5
5	Security / Authentication	Not yet implemented	Single seeded demo user; no login flow